

80 Ni-20 Cr:

$$H_t - H_0 = 0.1033t + 3.8(10^{-5})t^2 \quad (0^\circ \text{ to } 150^\circ \text{ C}). \quad (1)$$

$$H_t - H_0 = -0.33 + 0.1079t + 2.32(10^{-5})t^2 \quad (150^\circ \text{ to } 600^\circ \text{ C}). \quad (2)$$

$$H_t - H_0 = -4.37 + 0.1147t + 2.31(10^{-5})t^2 \quad (600^\circ \text{ to } 900^\circ \text{ C}). \quad (3)$$

Monel:

$$H_t - H_0 = 0.1010t + 2.20(10^{-5})t^2 \quad (0^\circ \text{ to } 300^\circ \text{ C}). \quad (4)$$

Stainless steel 347:

$$H_t - H_0 = 0.12508t + 1.703(10^{-5})t^2 - 9.31 \log_{10}[(t + 273.2)/273.2] \quad (0^\circ \text{ to } 900^\circ \text{ C}). \quad (5)$$

Stainless steel 446:

$$H_t - H_0 = 0.10781t + 5.37(10^{-5})t^2 \quad (0^\circ \text{ to } 500^\circ \text{ C}). \quad (6)$$

$$H_t - H_0 = -6.83 + 0.16178t - 123,000(10^{-0.0078t}) \quad (600^\circ \text{ to } 900^\circ \text{ C}). \quad (7)$$

Differentiation of these equations leads to the following equations for the specific heat (in cal g⁻¹ deg C⁻¹):

80 Ni-20 Cr:

$$C_p = 0.1033 + 7.6(10^{-5})t \quad (0^\circ \text{ to } 150^\circ \text{ C}). \quad (8)$$

$$C_p = 0.1079 + 4.64(10^{-5})t \quad (150^\circ \text{ to } 600^\circ \text{ C}). \quad (9)$$

$$C_p = 0.1147 + 4.62(10^{-5})t \quad (600^\circ \text{ to } 900^\circ \text{ C}). \quad (10)$$

Monel:

$$C_p = 0.1010 + 4.40(10^{-5})t \quad (0^\circ \text{ to } 300^\circ \text{ C}). \quad (11)$$

Stainless steel 347:

$$C_p = 0.1251 + 3.41(10^{-5})t - 4.04/(t + 273.2) \quad (0^\circ \text{ to } 900^\circ \text{ C}). \quad (12)$$

Stainless steel 446:

$$C_p = 0.1078 + 1.074(10^{-4})t \quad (0^\circ \text{ to } 500^\circ \text{ C}). \quad (13)$$

$$C_p = 0.1618 + 2209(10^{-0.0078t}) \quad (600^\circ \text{ to } 900^\circ \text{ C}). \quad (14)$$

Specific-heat values calculated for even temperatures from eq (8 to 14) are listed in table 3.

The specific heats of the four alloys are shown graphically in figures 1 to 4. The lines represent the smooth values given by eq (8 to 14), and the points correspond to the unsmoothed corrected values calculated for short temperature intervals from pairs of successive values given in table 2.

The representation of the specific heat of 80 Ni-20 Cr by straight lines in figure 1 is convenient and

TABLE 3. Specific heat (smoothed)

Temperature	Specific heat, cal g ⁻¹ deg C ⁻¹			
	80 Ni-20 Cr	Monel	Stainless steel—	
			347	446
° C				
0	0.1033	0.1009	0.1104	0.1078
25	.1052	.1021	.1123	.1105
50	.1071	.1033	.1142	.1132
100	.1109	.1054	.1176	.1185
200	.1171	.1097	.1233	.1293
300	.1217	.1142	.1283	.1401
400	.1264	-----	.1326	.1508
500	.1310	-----	.1370	.163
600	.14	-----	.1410	.21
700	.1470	-----	.1448	.1695
800	.1515	-----	.1487	.1631
900	.1563	-----	.1525	.1620

within the accuracy of the data, which however do not justify the assumption of a real discontinuity in slope at 150° C. On the other hand, real discontinuities do appear for 80 Ni-20 Cr near 600° C (fig. 1) and for stainless steel 446 between 500° and approximately 650° C (fig. 4). Although the total enthalpy changes over these regions were measured as precisely as in the other cases, the exact variation of specific heat with temperature in these regions of transition was not established because the enthalpy measurements did not cover sufficiently small temperature intervals. Otherwise the smoothed specific heats represented by eq (8 to 14), table 3, and the lines in figures 1 to 4 are believed to be accurate within ±2 percent, at least for the actual specimens measured.

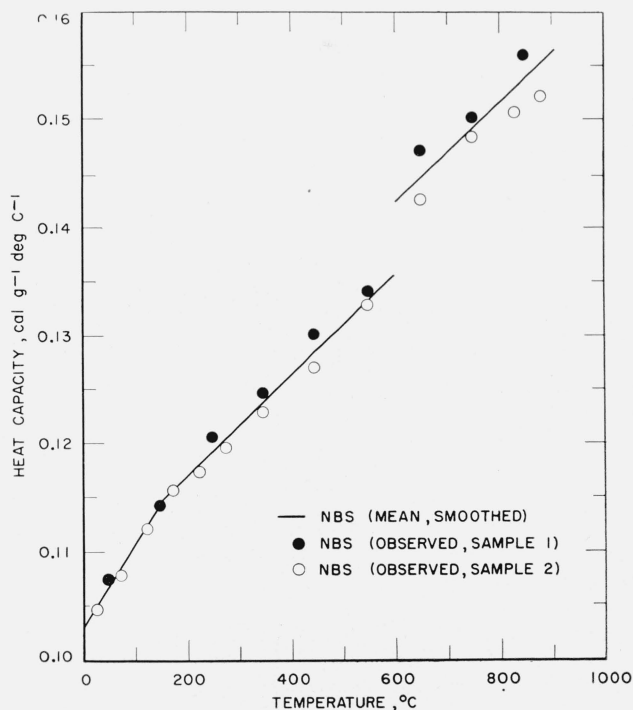


FIGURE 1. Specific heat of 80 Ni-20 Cr.